A-modules, and show that they are isomorphic to V^n and W^n, respectively. Deduce that V^n is isomorphic to W^n, and use the Krull-Schmidt theorem (valid over any field by Problem 3.8.3) to deduce that V is isomorphic to W.

(ii) (The Noether-Deuring theorem) In the setting of (i), suppose that $V \otimes_K L$ is a direct summand in $W \otimes_K L$ (i.e., $W \otimes_K L \cong V \otimes_K L \oplus Y$, where Y is a module over $A \otimes_K L$). Show that V is a direct summand in W.

Problem 3.8.5. Let A be the algebra of real-valued continuous functions on $\mathbb{R}$ which are periodic with period 1. Let M be the A-module of continuous functions f on $\mathbb{R}$ which are antiperiodic with period 1, i.e., $f(x+1) = -f(x)$.

(i) Show that A and M are indecomposable A-modules.

(ii) Show that A is not isomorphic to M but $A \oplus A$ is isomorphic to $M \oplus M$.

Remark 3.8.6. Thus, we see that, in general, the Krull-Schmidt theorem fails for infinite dimensional modules. However, it still holds for modules of **finite length**, i.e., modules M such that any filtration of M has length bounded above by a certain constant $l = l(M)$.

3.9. Problems

Problem 3.9.1. Extensions of representations. Let A be an algebra, and let V, W be a pair of representations of A. We would like to classify representations U of A such that V is a subrepresentation of U and $U/V = W$. Of course, there is an obvious example $U = V \oplus W$, but are there any others?

Suppose we have a representation U as above. As a vector space, it can be (nonuniquely) identified with $V \oplus W$, so that for any $a \in A$ the corresponding operator $\rho_U(a)$ has block triangular form

$$\rho_U(a) = \begin{pmatrix} \rho_V(a) & f(a) \\ 0 & \rho_W(a) \end{pmatrix},$$

where $f : A \to \mathrm{Hom}_k(W, V)$ is a linear map.

(a) What is the necessary and sufficient condition on $f(a)$ under which $\rho_U(a)$ is a representation? Maps f satisfying this condition are